

LP AND GAMES QHM5701– COURSEWORK

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This Coursework includes 4 problems. Please attempt all of them. The total number of marks for this Coursework is 45. You need to submit either a pdf or an image file on QMPlus.

Due date: December 5th 2025 at 23:59 Beijing time.

IMPORTANT. PLEASE READ.

- **All work submitted must be individual work.** You may discuss with friends, but the end work that you submit must be your own.
- Usage of technology is also considered similar to taking help from peers. Engaging with technology for assistance is acceptable but **plagiarism of any kind caught would be dealt with as misconduct.**
- **Steps taken and explanations to reach the result in every problem carries marks.** In all problems, final result alone **only** carries partial marks.

Problem 1. Let $P \subseteq \mathbb{R}^n$. We say P is a convex set if it satisfies the property that for all $x, y \in P$ we have $\lambda x + (1 - \lambda)y \in P$ for all $\lambda \in [0, 1]$ (i.e. for every x and y in P , every convex combination of x and y is also in P).

Consider an arbitrary linear programme in standard equation form. Prove that the set of all feasible solutions to this linear programme must be a convex set.

[5 marks]

Problem 2. Consider the following linear program:

$$\begin{aligned} \text{maximize} \quad & 120x_1 + 100x_2 \\ \text{subject to} \quad & 2x_1 + 2x_2 + x_3 = 8 \\ & 5x_1 + 3x_2 + x_4 = 15 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

- Convert the linear program into its matrix form.
- Determine a set of basic variables. Explain your answer.
- Determine a basic solution. Explain your answer.
- Find an extreme point solution. Explain your answer.

[8 marks]

Problem 3. For each of the following linear programs, sketch the constraints, restrictions and clearly shade the feasible region. Also, locate each optimal solution in your graph and explain how you found it using normal vectors. You need to create two plots.

The graph should be sketched by hand.

If you use any online graph tool or Python code, please clearly state it.

$$\begin{aligned}
 & \text{maximize} && 4x_1 + x_2 \\
 & \text{subject to} && 2x_1 + x_2 \leq 100 \\
 (1) & && x_1 + x_2 \leq 80 \\
 & && x_1 \leq 40 \\
 & && x_1, x_2 \geq 0.
 \end{aligned}$$

$$\begin{aligned}
 & \text{maximize} && 2x + 3y \\
 & \text{subject to} && x + y \leq 4 \\
 (2) & && 3x + y \leq 6 \\
 & && x + 3y \leq 6 \\
 & && x, y \geq 0.
 \end{aligned}$$

[6 marks]

Problem 4. As a sales representative at *BestMobileTech* store, you are in charge of arranging the sales of four different types of mobile phones. Each phone type provides a different profit, but the store faces limitations on three key resources: demonstration time, technical setup time, and display space.

Phone	Profit (RMB)	Demo (hours)	Setup (hours)	Display (units)
Premium (x_1)	5000	2	3	5
Mid-range (x_2)	2000	1	0	4
Budget (x_3)	1000	1	1	0
Business (x_4)	2000	2	0	1

You have a total demo time available of 6 hours per week, a total setup time available of 15 hours per week and a total display space available of 24 units. Your goal is to determine how many of each phone type to the store needs to focus on selling to maximise its total weekly income.

- Formulate this problem as a linear program and justify your answer.
- Find the maximum number of each phones which need to be sold in order to maximise the profit. Fully explain your answer.

[7 marks]

Problem 5. Find an optimal solution for each of the following linear programming problems using the simplex method. If there is no optimal solution or there are multiple optimal solutions, explain why. Justify any new variables you might introduce and all the steps you consider.

$$\begin{aligned}
 (3) \quad & \text{maximize} && 2x_1 + 3x_2 - x_3 \\
 & \text{subject to} && x_1 + 2x_2 - x_3 \leq 6 \\
 & && x_1 - 3x_2 - 3x_3 \leq 10 \\
 & && x_1, x_2, x_3 \geq 0
 \end{aligned}$$

$$\begin{aligned}
 (4) \quad & \text{maximize} && 3x_1 + 3x_2 - x_3 + x_4 \\
 & \text{subject to} && 2x_1 - x_2 - x_3 + x_4 \leq 2 \\
 & && x_1 - x_2 + x_3 - x_4 \leq 5 \\
 & && 3x_1 + x_2 + 5x_4 \leq 12 \\
 & && x_1, x_2, x_3, x_4 \geq 0
 \end{aligned}$$

$$\begin{aligned}
 (5) \quad & \text{maximize} && 64x_1 + 48x_2 + 8x_3 \\
 & \text{subject to} && 8x_1 + 6x_2 + x_3 \leq 48 \\
 & && 4x_1 + 2x_2 + 1.5x_3 \leq 20 \\
 & && 2x_1 + 1.5x_2 + 0.5x_3 \leq 8 \\
 & && x_2 \leq 5 \\
 & && x_1, x_2, x_3 \geq 0
 \end{aligned}$$

[12 marks]

Problem 6. Consider the following linear programming problem:

$$\begin{aligned}
 & \text{maximize} && 3x_1 + x_2 + 2x_3 \\
 & \text{subject to} && -x_1 + x_2 - x_3 = 1 \\
 & && x_1 + x_2 + 2x_3 \geq 3 \\
 & && 2x_1 + x_2 + 2x_3 \leq 4 \\
 & && x_1, x_2, x_3 \geq 0
 \end{aligned}$$

- (i) Convert the linear program into its standard equation form.
- (ii) Explain why **artificial variables** need to be introduced if we want to find an initial basic feasible solution of this problem.
- (iii) Construct the initial tableau of the II-phase simplex algorithm, which includes both objective functions $-w$ and $-z$ as shown during the lectures.
- (iv) “Clear” the tableau and bring it to a form where the II-phase simplex algorithm can be applied.

- (v) Solve the problem using the II-phase simplex algorithm by clearly mentioning the pivot steps and row operations. **You must not use any online or Python solver.**

[7 marks]

END OF COURSEWORK 1

TOTAL MARKS: 45

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